

FBD: FOVA IBD automation process:

Connect Data Science

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No headings included in this document

This document will intend to explain the logic and steps of the new tool in development for evaluation and proposal of the new FBD's. The approach used here may be transferible and use for other IBDs, such as DSASI and AFI IBDs. Using this tool the expectation is to end up with more homogenous IBD's, as consequence FOVA results will be more accurate

Data extraction:

The new tool that it will evaluate and propose improve version of FBD will require following input:

- Query that will extract the most relevant information for each store participating in FOVA validation for each country, will include by Existing FBD - Shop - Sales and # records of latest 12 weeks, plus #Items_commons, #items_new, #items_cancel, ... This query was prepared by Innovation IVSO Team and it is: [Input_prep_query.txt](#)
- Query that will extract the shores and madras basic Store_Chars used for cell definition: [Input_madras_chars_query.txt](#)¹

Automation method:

- The two Inputs are read and merged by shops_id
- Simple comparison of Char values between Madras_shop_chars and Sirval_Shop_chars from Store table
- Calculate the basic stats for existing FBDs, and cross check with FBD requirements (min 200 stores, CV_value < 0.5 & CV_ref < 0.3)
- If any or all existing FBD meets criteria those ones will stop the evaluation
- Check if any Channel_shoptypes have less than 60 stores and stores with no_shoptype in Madras, for those cases we will define FBD for each case, as nothing else can be done.
- Calculation of basic statistic at Channel Shoptype and Retailer level to check if any individual retailer may be a good candidate to FBD, if yes, define FBD for those cases
- For the remaining stores, we will count number of stores for each Channel-Shoptype, to define the max - number of splits we could do to have as homogenous as possible FBD
- Re-calculate basic stats (CV_value, CV_records, number of stores) for the following groups:
 - Channel-Shoptype (Level_0)
 - Channel-Shoptype-Retailer (Level_1)
- Considering those ones, we will split each retailer in the group as follows.
 - (df_work1.max_splits <= 1) - IBD – Those ones will be proposed as New FBD
 - (df_work1.max_splits == 2) & (df_work1.avg_val_l1 >= df_work1.median_val) – Large,
 - (df_work1.max_splits == 2) & (df_work1.avg_val_l1 < df_work1.median_val) – Small,

1. https://adlm.nielseniq.com/confluence/download/attachments/549749569/Input_madras_chars_query.txt?api=v2&modificationDate=1695908675397&version=1

- (df_work1.max_splits == 3) & (df_work1.avg_val_l1 >= df_work1.q75_val), – Large
- (df_work1.max_splits == 3) & (df_work1.avg_val_l1 > df_work1.q25_val) & (df_work1.avg_val_l1 < df_work1.q75_val), – Medium
- (df_work1.max_splits == 3) & (df_work1.avg_val_l1 <= df_work1.q25_val), – Small
- (df_work1.max_splits == 4) & (df_work1.avg_val_l1 >= df_work1.q75_val), – Large
- (df_work1.max_splits == 4) & (df_work1.avg_val_l1 > df_work1.median_val) & (df_work1.avg_val_l1 < df_work1.q75_val), – Large-Medium
- (df_work1.max_splits == 4) & (df_work1.avg_val_l1 > df_work1.q25_val) & (df_work1.avg_val_l1 < df_work1.median_val), – Small-Medium
- (df_work1.max_splits == 4) & (df_work1.avg_val_l1 <= df_work1.q25_val) – Small
- Re-evaluate if any of the group is less than 60 Stores, if yes merge with with closed one
- Check if these new groups build on shoctype retailer basis by sales range meet criteria expected criteria, if yes propose as new FBD
- Then it will require an evaluation for remainig groups that will be compared to observe if those ones are better than total shootypes, the ones with less CV combination will be proposed as New FBD
- Boxplot for new and old FBDs are also generated to final analysis.
- And finally, a report is generated that will contain:
 - List of Stores with Shoctype and Retailer values and new proposed FBD in each column
 - Final Stats for new and old FBDs are calculated and populated in output file (excel) one in each sheet.

Here you can see and example of the output file: [newfbd_PT_28-09-2023.xlsx](#)²

2. https://adlm.nielseniq.com/confluence/download/attachments/549749569/newfbd_PT_28-09-2023.xlsx?api=v2&modificationDate=1695908283657&version=1